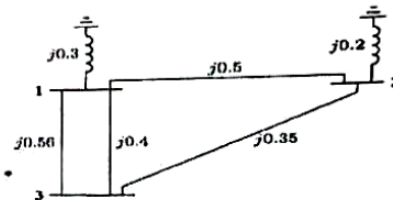


Time:- 50 Minutes

- Q.1: Derive the generalized expression of post fault bus voltage and line current in a n-bus large power system. [5]
Q.2: The bus impedance matrix for the network shown in figure is given below: [5]

$$Z_{bus} = \begin{bmatrix} j0.183 & j0.078 & j0.141 \\ j0.078 & j0.148 & j0.106 \\ j0.141 & j0.106 & j0.267 \end{bmatrix}$$



The line between buses 1 and 3 with impedance $Z_{bus} = j0.56$ is removed by the simultaneous opening of breakers at both ends of the line. Determine the new bus impedance matrix.

- Q.3: The system shown in figure below is initially on no load with generators operating at their rated voltage with their emfs in phase. The rating of the generators and the transformers and their respective percent reactances are marked on the diagram. All resistances are neglected. The line impedance is $j160$ ohms. A three-phase balanced fault occurs at the receiving end of the transmission line. Draw the reactance diagram and determine the short-circuit current. [5]

